Chiller



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Chiller

What it is used for

A chiller removes heat from water, a process fluid, or an enclosed space to maintain a specified temperature. Correct operating settings, clean heat-transfer surfaces, proper flow, and routine monitoring help maintain cooling performance and protect equipment and product quality.

Main parts

- Compressor raises refrigerant pressure in vapour-compression chillers.
- Condenser rejects heat from the refrigerant.
- Evaporator transfers heat from the process fluid or air.
- Expansion device regulates refrigerant flow into the evaporator.
- Pump circulates chilled water or process fluid where fitted.
- Temperature sensors monitor supply, return, and system conditions.
- Control panel sets operating parameters and displays alarms.
- Fans or cooling-water components remove heat from the condenser where fitted.
- Filters and strainers help prevent debris from entering fluid circuits.

Before you start

1. Only trained personnel should operate or service the chiller.
2. Read the equipment manual and confirm the intended application, setpoint, fluid type, flow requirements, and operating limits.
3. Inspect the housing, piping, hoses, valves, electrical connections, insulation, and visible refrigerant lines for damage or leakage.
4. Check fluid level, fluid concentration where applicable, and the condition of filters or strainers.
5. Ensure the condenser has adequate airflow or cooling-water supply and that ventilation paths are clear.
6. Verify that pumps, fans, guards, emergency stops, and alarms are available and in working condition.
7. Confirm that electrical supply and protective earth meet equipment requirements; refrigerant-system service must be performed by qualified personnel.

How to use

1. Place the chiller in its approved operating location with the required clearance and ventilation.
2. Check that the process circuit is filled, vented, and ready for operation according to the manual.
3. Open the required valves and confirm that the fluid circuit has no visible leaks.

4. Switch on the main supply and inspect the control panel for alarms before starting the cooling cycle.
5. Set the required temperature, flow conditions, and other permitted parameters according to the process specification.
6. Start the pump or circulation system in the sequence specified by the manufacturer.
7. Start the cooling cycle and verify that the compressor, fans, or cooling-water system operate as intended.
8. Monitor supply and return temperatures, fluid flow, pressure indicators, and alarm messages during operation.
9. Check the cooled process or product for the required temperature and record operating readings.
10. If abnormal noise, leakage, low flow, overheating, or a persistent alarm occurs, stop the unit safely and investigate using approved procedures.
11. At shutdown, follow the manufacturer's sequence for stopping cooling, pumps, and the main supply.
12. Isolate power before cleaning, opening panels, clearing filters, or performing maintenance.

⚠ Safety

- Do not bypass pressure switches, flow interlocks, guards, or other protective devices.
- Do not touch electrical components with wet hands or operate damaged electrical equipment.
- Refrigerant leaks may create toxicity, flammability, frostbite, or oxygen-displacement hazards depending on refrigerant type; follow the specified emergency procedure.
- Do not open a pressurised refrigerant circuit or attempt repairs without appropriate qualifications and equipment.
- Avoid contact with hot surfaces, rotating fans, and moving pump components.
- Use compatible process fluids and follow site procedures for chemical handling, spills, and disposal.

Care & storage

- Clean air filters, condenser surfaces, and strainers at the prescribed intervals.
- Inspect pumps, fans, valves, hoses, insulation, and visible pipework for wear or leaks.
- Check fluid levels and quality according to the manufacturer's schedule.
- Verify temperature sensors, alarms, and control settings at the recommended service intervals.
- Arrange qualified maintenance for refrigerant-system checks and investigate recurring faults promptly.

Common mistakes to avoid

- Operating with insufficient fluid flow or an incorrectly filled circuit.
- Blocking condenser airflow or neglecting filter cleaning.
- Setting a temperature outside the equipment's rated operating range.
- Ignoring alarms, refrigerant leakage signs, or repeated compressor trips.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.